

Additional Admissions Form

Applicant report · Academic year 2025-2026



Applicant

Full name	Ivayla Ilieva
TU/e student number	1234567

Previous studies

Degree 1: Bachelor Computer Science and Engineering

Degree programme	Bachelor Computer Science and Engineering
Graduation date	22 Aug 2026
University	Technical university of Eindhoven
Location	Eindhoven, The Netherlands
Total degree credits	180

Relevant courses

Course 1: 2MBA20 Linear algebra

Degree	Bachelor Computer Science and Engineering — Technical university of Eindhoven
Course credits	5 / 180 ECTS
Final grade	9
Official course description	Systems of linear equations, matrices and vector spaces are the core topics in this course. Systems of linear equations and matrix arithmetic can be seen as the computational vehicles for working in vector spaces, but are also interesting in themselves and used everywhere in science. Vector spaces are a convenient way of mathematically modelling and working with space (and to generalize!) and elementary vector space theory mostly focuses on linear aspects. Vector spaces consist of vectors (just think of arrows with a certain direction and a certain length, arrows that can be added and multiplied by scalars - that's the algebra!) and provide a way to define and work with notions like distance, angle, dimension, linear objects, maps between linear objects.

Course 2: 2IAB1 Foundations of Data Analytics

Degree	Bachelor Computer Science and Engineering – Technical university of Eindhoven
Course credits	5 / 180 ECTS
Final grade	8.5
Official course description	General learning goals Use basic statistical concepts and techniques and perform appropriate statistical tests Choose and apply suitable visualization techniques Analyze and model data using linear regression, clustering, decision tree mining and association rules learning Read and make simple database schemes and simple queries to a database. Clean data, choose and apply data transformations, data reduction, and data discretization Understand, interpret, and document data and information in the context of realistic scenarios, structure open problems along the phases of the data life cycle, formulate hypotheses, incorporate ethical considerations and reflect on analysis strengths and limitations Use tools for implementing data engineering tasks (Python with Jupiter Notebooks) in a structured way Choose and communicate interesting findings in the language understandable for their end user (visually or textually).

Prerequisite coverage

Overview

1. Linear Algebra	Covered (4 of 4)
2. Logic	Not covered
3. Discrete Mathematics	Not covered
4. Probability and Statistics	Not covered
5. Data Structures and Algorithms	Not covered
6. Object-Oriented Programming and Software Development	Not covered
7. Data Modeling and Databases	Not covered
8. Machine Learning and Data Mining	Not covered
9. Visualization	Partially covered (1 of 3)
10. Group Project Work	Not covered

1. Linear Algebra

Required knowledge	Covered by
Calculate with matrices and vectors	Course 1 · 5 ECTS · grade 9
Solve linear systems with Gauss(-Jordan) elimination	Course 1 · 5 ECTS · grade 9
Rank, orthogonality, independence and eigenvectors	Course 1 · 5 ECTS · grade 9
Implement linear algebra calculations	Course 1 · 5 ECTS · grade 9

2. Logic

Required knowledge	Covered by
Propositional logic and predicate logic	—
Reason with logical formulas	—
Provide proofs using various proving techniques (including induction, case distinction and contradiction)	—

No relevant course selected.

3. Discrete Mathematics

Required knowledge	Covered by
Concepts and techniques from discrete mathematics (set theory, functions, relations and orderings)	—
Basic graph theory (graphs, trees and their properties)	—
Use discrete mathematics in computations and formal reasoning	—

No relevant course selected.

4. Probability and Statistics

Required knowledge	Covered by
Introductory probability theory and discrete and continuous random variables	—
Descriptive statistics, confidence intervals, hypothesis testing and estimation theory	—

No relevant course selected.

5. Data Structures and Algorithms

Required knowledge	Covered by
Algorithm design techniques and standard data structures	—
Prove correctness of algorithms and reason about algorithm complexity	—

No relevant course selected.

6. Object-Oriented Programming and Software Development

Required knowledge	Covered by
Write programs from scratch in imperative or object-oriented languages	—
Use general algorithmic techniques (aggregation, searching, sorting and recursion)	—
Apply principles of code quality and software engineering	—

No relevant course selected.

7. Data Modeling and Databases

Required knowledge	Covered by
Design data models (E-R diagrams and UML) from natural-language requirements	—
Query relational databases based on natural-language requirements	—

No relevant course selected.

8. Machine Learning and Data Mining

Required knowledge	Covered by
Theoretical foundations of data mining and machine learning	—
Apply feature selection and extraction	—
Apply supervised learning (classification and regression)	—
Apply unsupervised learning (clustering and matrix factorization)	—
Apply evaluation methods and understand overfitting	—

No relevant course selected.

9. Visualization

Required knowledge	Covered by
Basic principles of visualization	Course 2 · 5 ECTS · grade 8.5
Design, implement and evaluate visualization tools	—
Implement data transformation and visualizations using visual variables and interaction principles	—

Additional explanation — Course 2

One of the learning goals of the course was "Choose and apply suitable visualization techniques" and I believe it covers the selected topic.

10. Group Project Work

Required knowledge	Covered by
Presenting and writing skills in group project work	—

No relevant course selected.

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